

Chapter 32 - Coprocessor and Cross-CPU Calls

The six processors of Part IV sit on the same Intuition Engine bus. They see shared RAM, they can reach the same cards, and they can pass work to each other without changing machines. A BASIC program running on IE64 can submit work to a 6502 service, and another CPU can poll for the answer later.

The mechanism is the **coprocessor system**. It consists of MMIO registers, one ring buffer per worker instance, completion records, and eight BASIC forms and functions: `COSTART`, `COSTOP`, `COCALL`, `COSTATUS`, `COWAIT`, `COCAPS`, `COINSTANCE`, and `COSELSTATE`.

The reader path in this chapter is BASIC first, then direct `POKE` and `PEEK` of the same registers. The runnable example does not require a service program to be present; it proves the status and error path that every real service call uses.

32.1 Model

A **coprocessor** is one of the six CPUs acting as a worker for another CPU. The calling CPU submits a request descriptor; the worker reads it, writes a response descriptor, and marks the ticket complete.

The normal sequence is:

1. Start a worker with `COSTART`.
2. Put request bytes in shared RAM.
3. Call `COCALL` to enqueue the request.
4. Use `COSTATUS` to poll, or `COWAIT` to wait.
5. Read the response bytes.
6. Stop the worker with `COSTOP` when it is no longer needed.

Worker state is reported through `COPROC_WORKER_STATE`:

State bit	Meaning
Clear	No worker is running for that CPU type.
Set	At least one instance of that CPU type is running.

`COPROC_WORKER_STATE` aggregates instances: the bit is set when any instance of that type is running. To test a specific instance, select it with `COPROC_CPU_TYPE` and `COPROC_INSTANCE`, then read `COPROC_SELECTED_STATE`. M68K, x86, and IE64 support instances 0 and 1; IE32, 6502, and Z80 support instance 0 only.

Failures are always visible through `COPROC_CMD_STATUS` and `COPROC_CMD_ERROR`. `COCALL` also returns ticket 0 when enqueue fails, so a BASIC programme can reject the request immediately.

32.2 CPU Type Codes

These are the CPU type values used by BASIC and MMIO:

CPU type	Code	Service suffix	Instances	Worker RAM
IE32	1	. IE32	0	\$200000 to \$27FFFF
IE64	2	. IE64	0, 1	\$3A0000 to \$41FFFF; \$520000 to \$59FFFF

CPU type	Code	Service suffix	Instances	Worker RAM
6502	3	.IE65	0	\$300000 to \$30FFFF
M68K	4	.IE68	0, 1	\$280000 to \$2FFFFFF; \$420000 to \$49FFFF
Z80	5	.IE80	0	\$310000 to \$31FFFF
x86	6	.IE86	0, 1	\$320000 to \$39FFFF; \$4A0000 to \$51FFFF

COSTART checks that the service suffix matches the requested CPU type. A mismatch reports an error rather than starting the wrong worker.

32.3 BASIC Entry Points

32.3.1 COSTART

```
COSTART cpuType, "serviceName"
COSTART cpuType, instance, "serviceName"
```

Starts the named service on the selected CPU. The service must be available in IE storage and must match the CPU type. The optional instance selects a worker instance (default 0); M68K, x86, and IE64 accept 0 or 1, while the others accept only 0. The command records success or failure in COPROC_CMD_STATUS and COPROC_CMD_ERROR.

32.3.2 COSTOP

```
COSTOP cpuType
COSTOP cpuType, instance
```

Stops the worker for cpuType and optional instance (default 0). Requests that have not completed are marked COPROC_TICKET_WORKER_DOWN.

32.3.3 COCALL

```
ticket = COCALL(cpuType, op, reqPtr, reqLen, respPtr, respCap)
ticket = COCALL(cpuType, instance, op, reqPtr, reqLen, respPtr, respCap)
```

Enqueues a request. The seven-argument form selects a worker instance (default 0 in the six-argument form). The request is reqLen bytes starting at reqPtr; the response can use up to respCap bytes at respPtr. The op value is a service-defined operation code. A malformed argument count raises ?FC ERROR.

For any non-zero reqLen or respCap, the corresponding pointer must be a public low32 span allocated by MEMALLOC. Invalid raw pointers raise ?FC ERROR. Zero-length request or response spans may use pointer 0.

CCALL returns a non-zero ticket on success. It returns 0 on enqueue failure; read COPROC_CMD_ERROR to learn why.

Think of the ticket as a bus receipt. The caller owns the request bytes, the worker owns the service code, and the coprocessor block is the clerk between them. That model is the same whether the worker plays audio, fills a table, prepares a sprite, or converts data for another chip.

32.3.4 COSTATUS

```
status = COSTATUS(ticket)
```

Polls a ticket without waiting:

Value	Constant	Meaning
0	COPROC_TICKET_PENDING	Queued, not yet started
1	COPROC_TICKET_RUNNING	Worker is processing
2	COPROC_TICKET_OK	Completed successfully
3	COPROC_TICKET_ERROR	Worker returned an error
4	COPROC_TICKET_TIMEOUT	Wait deadline expired
5	COPROC_TICKET_WORKER_DOWN	Worker is no longer running

32.3.5 COWAIT

```
COWAIT ticket [, timeoutMs]
```

Waits for a ticket to finish or for the timeout to expire. If the timeout is omitted, BASIC uses 1000 milliseconds. COWAIT is a statement, not a function; call COSTATUS(ticket) afterwards to read the final ticket status.

32.3.6 COCAPS, COINSTANCE, COSELSTATE

```
limit = COCAPS(cpuType)
inst   = COINSTANCE()
state  = COSELSTATE()
```

COCAPS(cpuType) returns the number of worker instances a CPU type supports: 2 for M68K, x86, and IE64, 1 for the others. COCAPS also selects cpuType for the following COSELSTATE read. COINSTANCE() returns the currently selected worker instance. COSELSTATE() returns the running state of the selected (cpuType, instance) worker: 1 if online, 0 otherwise. These three functions work in direct mode and stored interpreted programmes, like COCALL and COSTATUS; they cannot be used in an AOT-compiled programme.

32.4 Runnable Status Example

This listing intentionally submits a request when no 6502 worker is running. It is useful because it exercises the same command, ticket, and error registers that a real service call uses, without requiring a service program first.

Type this listing:

```

10 REM PUT REQUEST AND RESPONSE BUFFERS IN SHARED RAM
20 REQ=MEMALLOC(8,4):RESP=MEMALLOC(16,4)
30 POKE32 REQ,123
40 REM ASK CPU TYPE 3, THE 6502, TO RUN OPERATION 1
50 T=COCALL(3,1,REQ,4,RESP,4)
60 PRINT "TICKET ";T
70 PRINT "CMD ";PEEK32(&H000F2348)
80 PRINT "ERR ";PEEK32(&H000F234C)
90 PRINT "WORKERS ";PEEK32(&H000F2374)

```

Expected result:

```

TICKET 0
CMD 1
ERR 6
WORKERS 0

```

CMD 1 means command error. ERR 6 is COPROC_ERR_NO_WORKER. Once a service is running, the same COCALL form returns a non-zero ticket instead.

Lines 20 and 30 create the same request buffer a real worker would read. Line 50 is the important control write hidden behind the BASIC function: it enqueues a request for CPU type 3. The three PEEK lines then show the command status, exact error code, and live worker mask in the MMIO block.

32.5 MMIO Register Block

The coprocessor block occupies \$F2340 to \$F238F. The extended status block occupies \$F23B0 to \$F23BF, and the capability and version-discovery block occupies \$F25A0 to \$F25BF.

Address	Name	Access	Purpose
\$F2340	COPROC_CMD	W	Command to run
\$F2344	COPROC_CPU_TYPE	W	Target CPU type
\$F2348	COPROC_CMD_STATUS	R	0 ok, 1 error
\$F234C	COPROC_CMD_ERROR	R	Error code
\$F2350	COPROC_TICKET	R/W	Ticket ID
\$F2354	COPROC_TICKET_STATUS	R	Last-pollled ticket status
\$F2358	COPROC_OP	W	Service operation code
\$F235C	COPROC_REQ_PTR	W	Request buffer pointer
\$F2360	COPROC_REQ_LEN	W	Request buffer length
\$F2364	COPROC_RESP_PTR	W	Response buffer pointer
\$F2368	COPROC_RESP_CAP	W	Response buffer capacity
\$F236C	COPROC_TIMEOUT	W	Timeout in milliseconds
\$F2370	COPROC_NAME_PTR	W	Pointer to service-name string
\$F2374	COPROC_WORKER_STATE	R	Per-type bitmask: any instance running
\$F2378	COPROC_STATS_OPS	R	Total operations dispatched

Address	Name	Access	Purpose
\$F237C	COPROC_STATS_BYTES	R	Total bytes processed
\$F2380	COPROC_IRQ_CTRL	R/W	bit 0 enables IRQ on completion
\$F2384	COPROC_DISPATCH_OVERHEAD	R	Last dispatch overhead, nanoseconds
\$F2388	COPROC_COMPLETED_TICKET	R	Last completed ticket ID
\$F238C	COPROC_INSTANCE	R/W	Worker instance selector (0 = default)
\$F23B0	COPROC_RING_DEPTH	R	Selected CPU ring occupancy
\$F23B4	COPROC_WORKER_UPTIME	R	Selected worker uptime, seconds
\$F23B8	COPROC_STATS_RESET	W	Write 1 to clear statistics
\$F23BC	COPROC_BUSY_PCT	R	Rolling worker busy percentage
\$F25A0	COPROC_INSTANCE_LIMIT	R	Instance count for COPROC_CPU_TYPE
\$F25A4	COPROC_SELECTED_STATE	R	Selected (<i>type</i> , <i>instance</i>) running state
\$F25A8	COPROC_MAILBOX_VERSION	R	Mailbox layout version
\$F25AC	COPROC_WORKER_BASE	R	Selected worker window base address
\$F25B0	COPROC_WORKER_END	R	Selected worker window end address
\$F25B4	COPROC_WORKER_RING	R	Selected worker ring base address
\$F25B8	COPROC_INSTANCE_STATE	R	Per-(<i>cpuType</i> , <i>instance</i>) running bitmask

The capability block (\$F25A0 and above) is reachable by the 32-bit addressing CPUs and the main-CPU BASIC runtime, not by the 6502 or Z80 \$F200 gateway. A coprocessor service running on those cores learns the mailbox layout version from its ring header rather than this block.

Read COPROC_RING_DEPTH and COPROC_WORKER_UPTIME only after writing both COPROC_CPU_TYPE and COPROC_INSTANCE; together the selectors choose which worker is queried.

COPROC_BUSY_PCT is calculated over the recent coprocessor activity window, about one second. Writing 1 to COPROC_STATS_RESET clears the operation and byte counters and starts a fresh busy-percentage window.

The 6502 and Z80 cannot address \$F2340 directly. They use a gateway at \$F200 to \$F24F. The gateway keeps the same register offsets, so \$F204 reaches COPROC_CPU_TYPE and \$F210 reaches COPROC_TICKET. Byte writes compose 32-bit values in little-endian order.

32.6 Commands

Write input registers first, then write one of these values to COPROC_CMD:

Code	Constant	Effect
1	COPROC_CMD_START	Start a worker
2	COPROC_CMD_STOP	Stop a worker
3	COPROC_CMD_ENQUEUE	Submit a request and return a ticket
4	COPROC_CMD_POLL	Poll a ticket's status
5	COPROC_CMD_WAIT	Wait for ticket completion or timeout

Code	Constant	Effect
6	COPROC_CMD_START_MEM	Start a worker from a guest-RAM image

After the command write, read COPROC_CMD_STATUS. If it is 1, read COPROC_CMD_ERROR.

COPROC_CMD_START_MEM uses the selected COPROC_CPU_TYPE and reads the service image from guest RAM at COPROC_REQ_PTR for COPROC_REQ_LEN bytes. It is the raw MMIO form used by self-contained programmes that already carry their worker image in memory. Ordinary BASIC programmes normally use CSTART for named worker files.

32.7 Error Codes

Code	Constant	Meaning
0	COPROC_ERR_NONE	Success
1	COPROC_ERR_INVALID_CPU	COPROC_CPU_TYPE is not in 1 to 6
2	COPROC_ERR_NOT_FOUND	Service was not found
3	COPROC_ERR_PATH_INVALID	Service name is malformed
4	COPROC_ERR_LOAD_FAILED	Service could not be loaded
5	COPROC_ERR_QUEUE_FULL	Worker's ring buffer has no free slots
6	COPROC_ERR_NO_WORKER	No worker is running for this CPU type
7	COPROC_ERR_STALE_TICKET	Ticket has already been reaped
8	COPROC_ERR_INVALID_INSTANCE	CPU type is valid but the instance is beyond its limit
9	COPROC_ERR_STALE_WORKER	Worker did not acknowledge the mailbox layout version

32.8 Ring Buffer Layout

Each worker has a 16-entry ring in shared RAM. The mailbox holds twelve ring slots at a uniform \$400 stride starting at MAILBOX_BASE (\$790000), ending at \$792FFF. The ring index is $\text{cpuTypeIndex} * 2 + \text{instance}$: instance 0 of each type uses the even ring, instance 1 the following odd ring. The three single-instance types (IE32, 6502, Z80) leave their odd rings reserved but unused. Byte offset \$04 of each ring header holds the worker's echoed layout version; a conforming worker copies the layout byte at +\$03 to +\$04 at startup or the manager refuses to route it work.

Item	Address or size
Mailbox base	\$790000
Ring base	$\$790000 + (\text{cpuTypeIndex} * 2 + \text{instance}) * \400
Ring header	head at +\$00, tail at +\$01, capacity at +\$02
Version gate	layout version at +\$03, worker acknowledgement at +\$04
Request area	16 request descriptors, 32 bytes each
Response area	16 response descriptors, 16 bytes each

The caller writes all request fields and the pending response descriptor before advancing head. The new head value publishes the completed request slot to the worker. The worker writes the response data, length, result, and final status

before advancing `tail`. The new tail value publishes the completed response to the caller. Do not advance either index early.

The CPU type indices are IE32 0, 6502 1, M68K 2, Z80 3, x86 4, and IE64 5. A worker must copy byte `+$03` to `+$04` during startup. If it does not acknowledge the current layout within the start gate, the command fails with error 9 and no requests are routed to it.

M68K receives its assigned ring base in A4, x86 in EBP, and IE64 in R30. This lets either instance use the same service image. IE32, 6502, and Z80 have one fixed ring each.

Request descriptor words:

Offset	Meaning
\$00	Ticket
\$04	CPU type
\$08	Operation code
\$0C	Timeout
\$10	Request pointer
\$14	Request length
\$18	Response pointer
\$1C	Response capacity

Response descriptor words:

Offset	Meaning
\$00	Ticket
\$04	Ticket status
\$08	Service result code
\$0C	Response length

Most BASIC programmes never touch these descriptors directly. They are included so machine-code services can be written and checked by hand in IE Mon.

32.9 Positive Service Pattern

When a service is present, the BASIC side looks like this:

```
10 REQ=MEMALLOC(8,4)
20 RESP=MEMALLOC(16,4)
30 POKE32 REQ,42
40 COSTART 3,"MUL.IE65"
50 T=COCALL(3,1,REQ,4,RESP,4)
60 IF T=0 THEN PRINT "SUBMIT FAILED":END
70 COWAIT T,1000
80 S=COSTATUS(T)
90 IF S<>2 THEN PRINT "STATUS ";S:END
100 PRINT "ANSWER ";PEEK32(RESP)
110 COSTOP 3
```

The service defines operation 1, reads the request word at REQ, writes a result word at RESP, then marks the ticket COPROC_TICKET_OK. The calling side does not need to know which instructions the worker used; it only needs the request and response format.

32.10 IRQ on Completion

Bit 0 of COPROC_IRQ_CTRL enables an interrupt when any ticket completes. The interrupt vector depends on the listening CPU; see Chapter 31, section 31.3. The handler reads COPROC_COMPLETED_TICKET to learn which ticket finished.

This is useful when the listening CPU has other work to do and does not want to block in COWAIT. The usual pattern is to submit several requests, return to other work, and let the completion interrupt wake up the response handler.

32.11 What Comes Next

Chapter 33 covers IE Mon, the interactive monitor that lets you single-step any CPU, examine registers, set breakpoints and watchpoints, and inspect the bus from a command prompt. Chapter 34 covers IE Script, the scripting surface that drives the monitor from BASIC-like programs.